

Powertrain solutions

Electric Machines

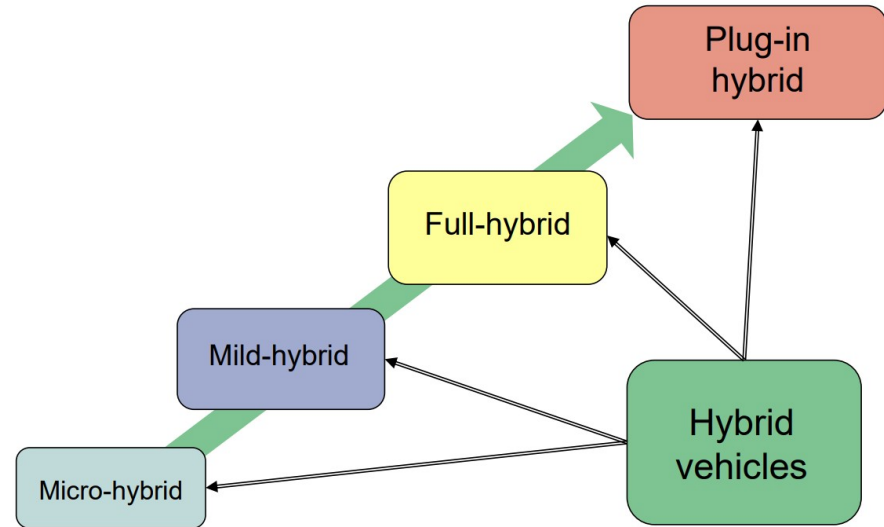


Powertrain Architecture



Hybrid vehicles Solutions

- **Mild Hybrid Electric Vehicle (MHEV):** Vehicle driven by a combustion engine powered by a compact electric motor. It can have the electric motor in different positions.
- **Full Hybrid Electric Vehicle (FHEV):** It has both combustion and electric engines. At low speeds and short distances, a fully electric propulsion is possible.
- **Plug-in Hybrid Electric Vehicle (PHEV):** It reaches higher speeds and longer distances unlike the Full Hybrid. They can only be charged from external points and can have a larger battery pack than the FHEV, because the size is not limited to the stored kinetic energy.





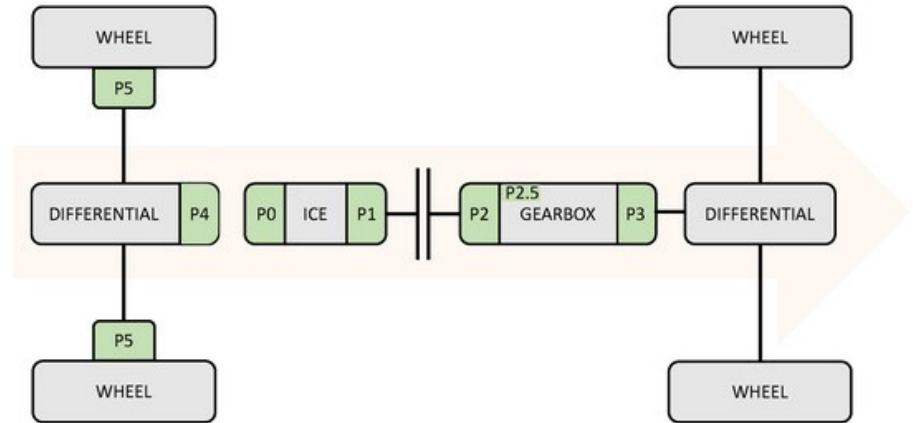
P0, P1, P2, P3, P4, P5

Hybrid vehicles use an internal combustion engine (ICE) and at least one electric motor (eMotor or EM).

In hybrid electric vehicles we can find many abbreviations as P0, P1, P2, P3, P4 and P5. Each abbreviation depends on the position of the electric motor in the hybrid powertrain.

In some way, the number identifies the distance between the electric motor and the wheels, which decreases from P0 to P5.

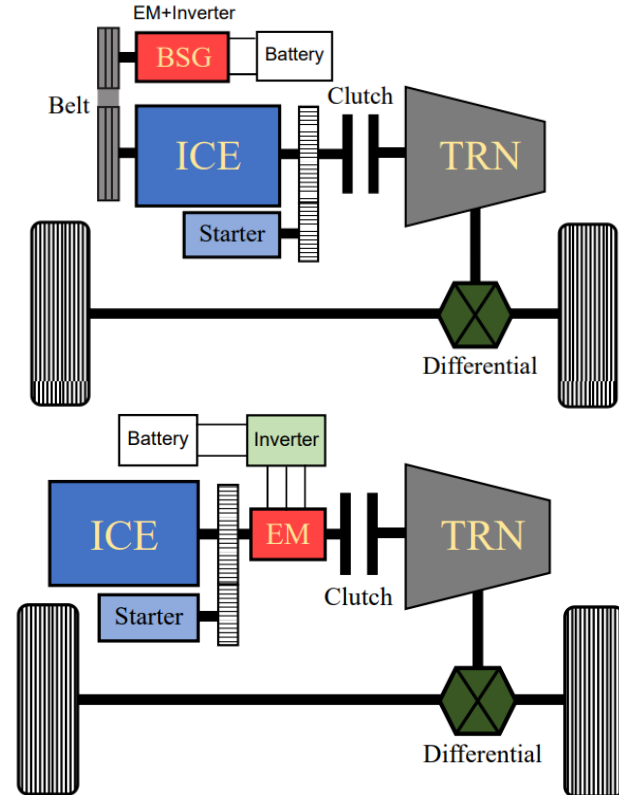
Hybrid vehicles





Hybrid vehicles P0 and P1

- **P0 configuration:** The EM can be mounted in place of the alternator and then connected to the engine with a belt. These are Belt-driven Starter Generators (BSG) which also act as a starter motor and generator.
- **P1 configuration:** The EM is inserted on the motor shaft. It runs at the same speed as the crankshaft and requires some more modifications.





Hybrid vehicles P0 and P1

- **Audi System** (P0 configuration)

- 1) DC-DC converter
- 2) 12V battery
- 3) 48V battery
- 4) Integrated BSG
- 5) ICE

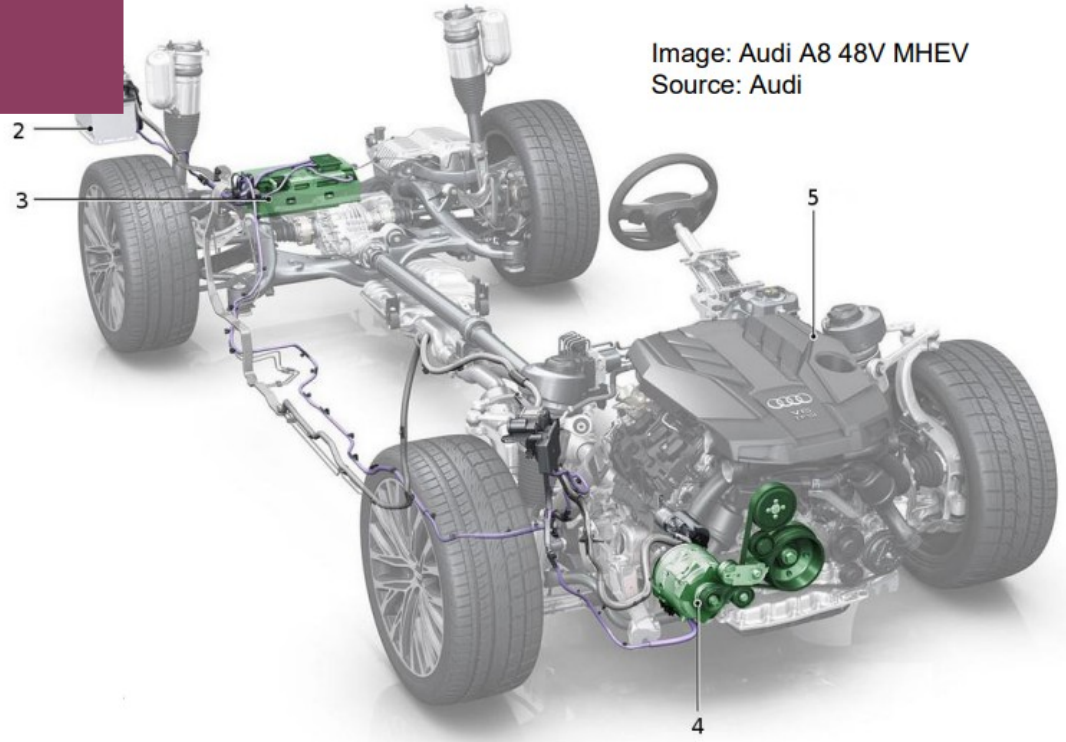
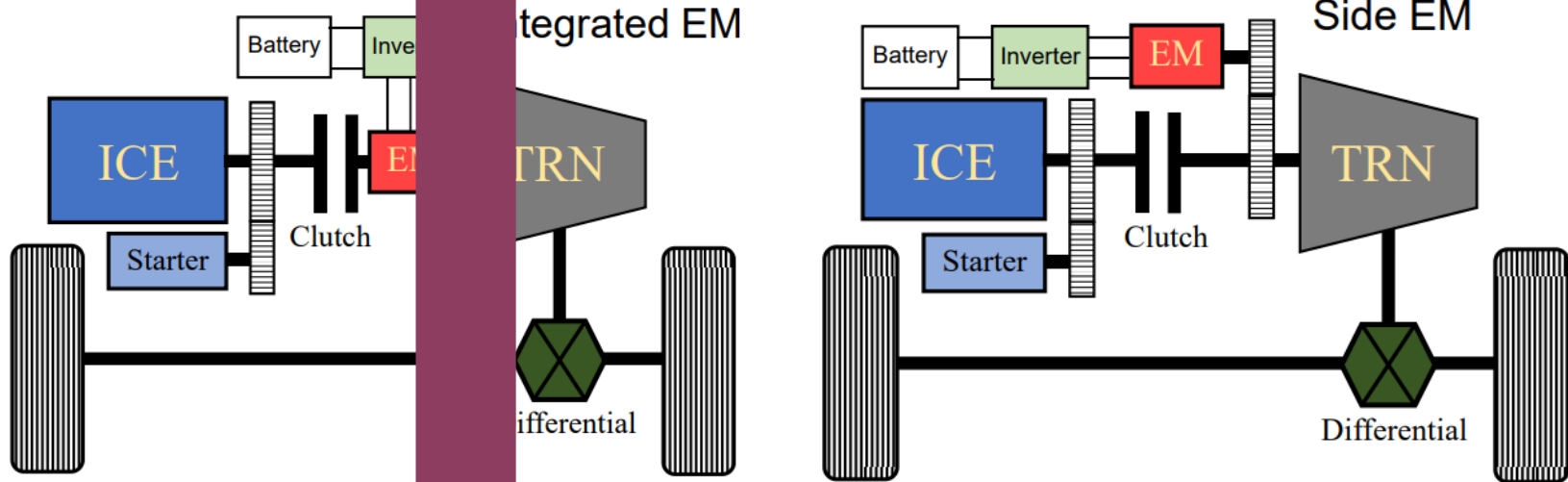


Image: Audi A8 48V MHEV
Source: Audi



Hybrid vehicles P2

- **P2 configuration:** The electric motor is placed between the engine and the transmission, directly on the shaft or through a gearbox. The engine insertion system implies a clutch that disconnects the ICE. So it is possible to drive in full electric by the opening of the clutch.

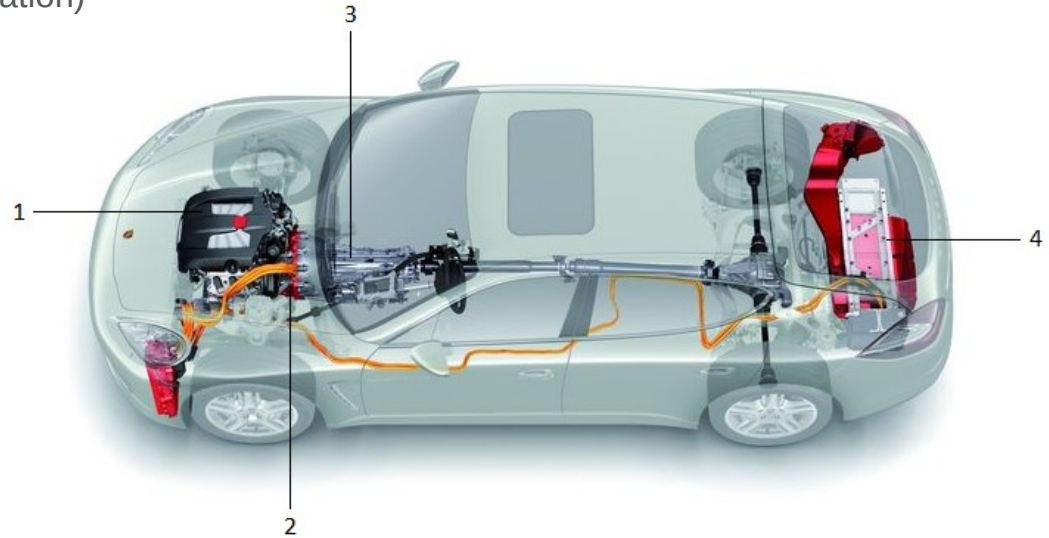




Hybrid vehicles P2

- **Porsche Panamera System (configuration)**

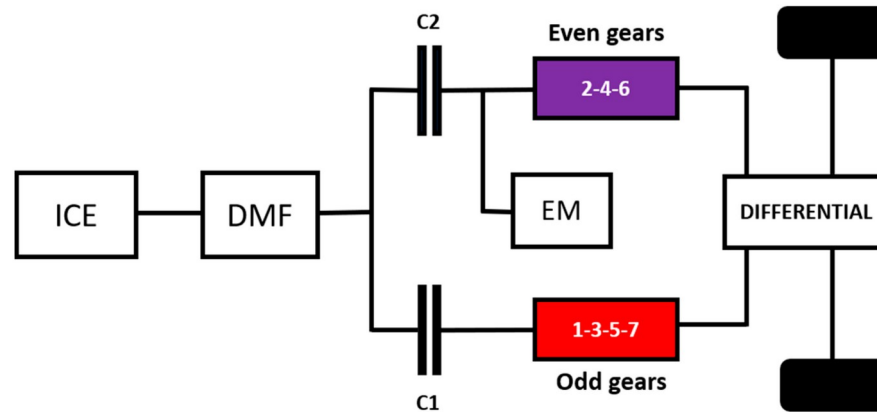
- 1) ICE
- 2) EM
- 3) ZF Transmission
- 4) Battery pack





Hybrid vehicles P2.5

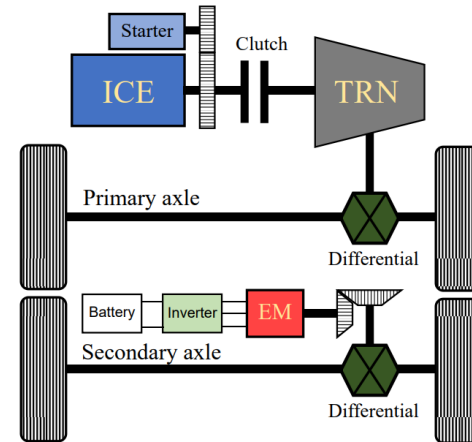
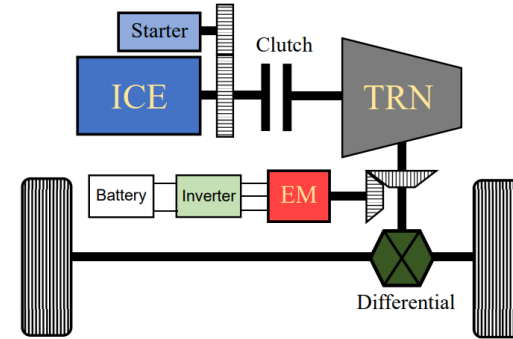
- **P2.5 configuration:** This configuration is possible only with a Dual Clutch Transmission. The electric motor is placed **inside** the transmission, on one of the two concentric transmission shafts, so that one shaft (the one of the odd gears) is completely driven by the ICE, while the other one (even gears) is driven also by the electric motor. Electric creeping/drive is possible by moving just the even gear shaft, but also combined combustion and electric power mode is possible.





Hybrid vehicles P3 and P4

- **P3 configuration:** This configuration moves the engine “downstream” of the transmission. In this way, losses are minimized, since the electric motor only drives the final part of the transmission.
- **P4 configuration:** The EM acts exclusively on the axis not connected to the ICE. In this way it is possible to have an all-wheel drive without mechanical connection with the thermal engine. Even the pure electric mode is easily obtainable, by disconnecting the main transmission.

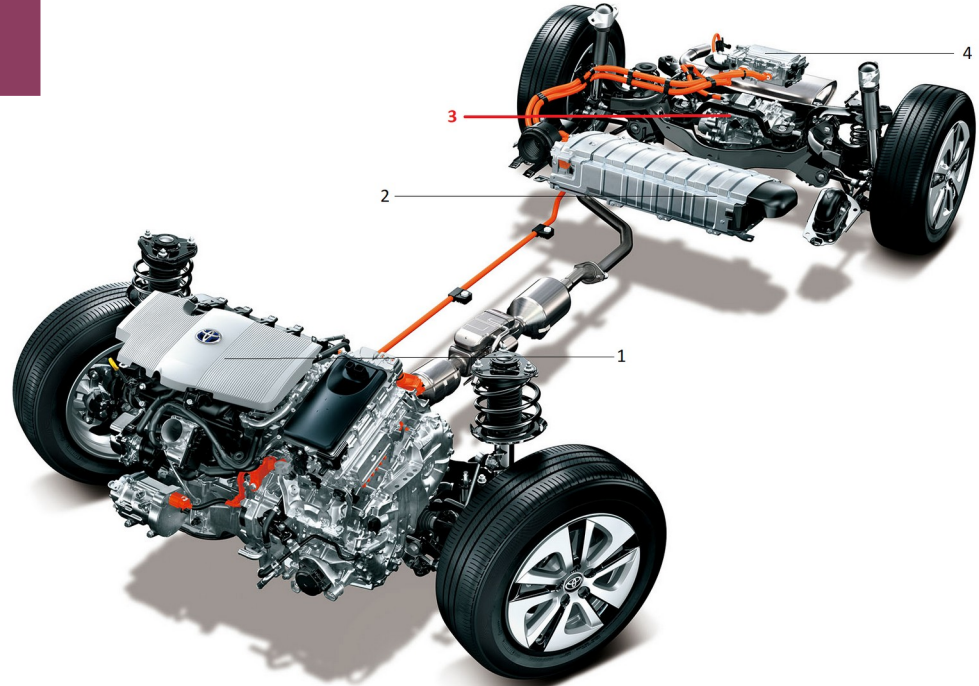




Hybrid vehicles P3 and P4

- **Toyota Prius IV (P4 config**

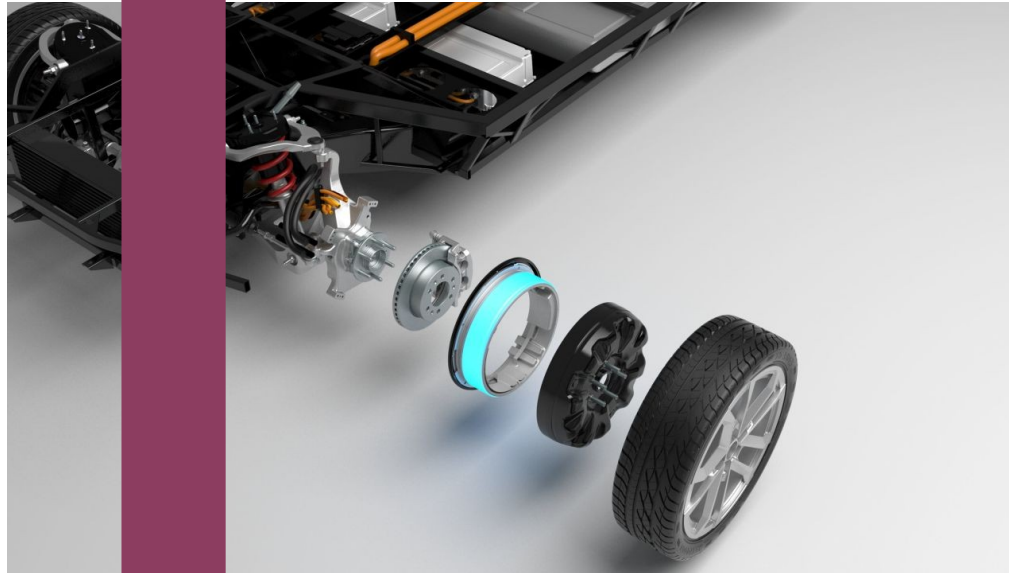
- 1) ICE
- 2) Battery pack
- 3) Induction EM
- 4) Inverter





Hybrid vehicles P5

- **P5 configuration:** For hybrid and full electric vehicles there is the acronym P5 which indicates the motors in the wheels, a solution that eliminates the "distance". This abbreviation, unlike those P0, P1, P2, P3 and P4, has a still limited application in production cars.





Hybrid vehicles

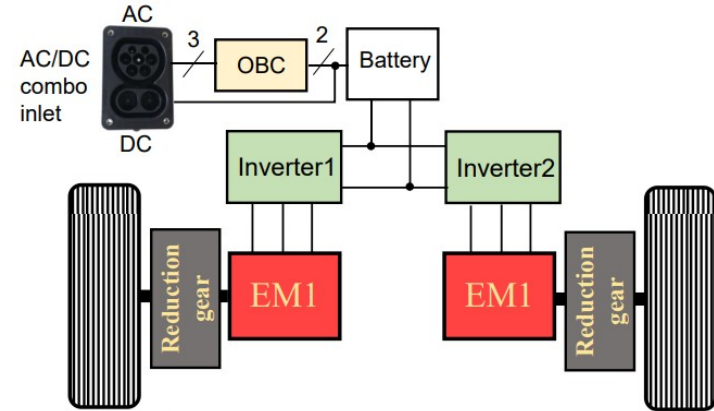
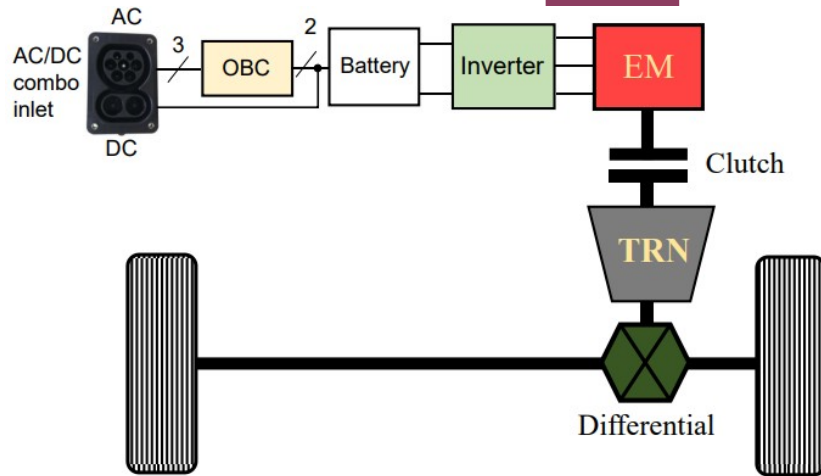
P1+P2 and P1+P4 configurations

- **P1 + P2 and P1 + P4 configurations:** P2 to P4 configurations as said have the electric motors placed after the clutch, connected to the driveline side, and that, can improve overall electrical performances; one of the most useful functionality of recent ICE vehicles is inhibited: S&S. To avoid this, another electric motor (a P1) can be added to the engine side enabling S&S functionality and enlarging hybrid strategies in terms of load shift, torque assist, regeneration etc... On the other hand, complexity and costs increase exponentially due to the addition of another motor that should work together all the other components. This configuration essentially combines both benefits and disadvantages of P1, P2 and P4s.



Full electric vehicles BEV

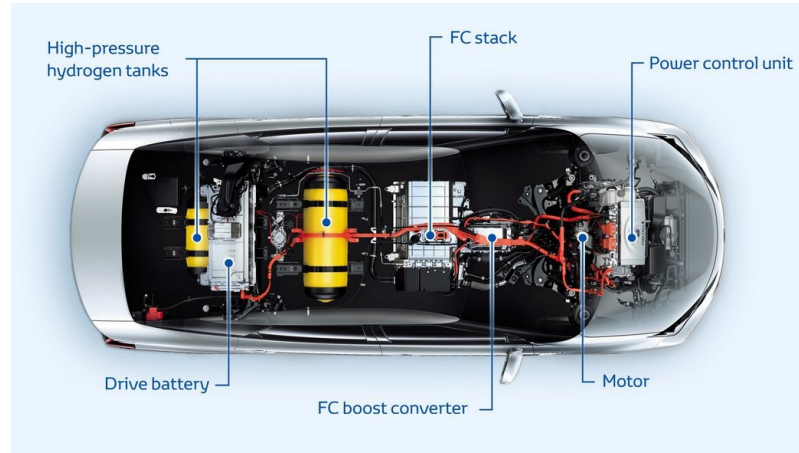
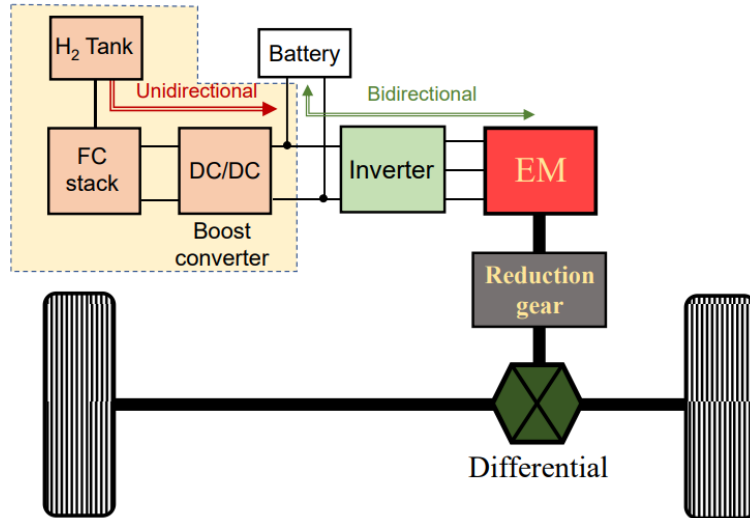
- **Battery Electric Vehicle (BEV):** The powertrain of an EV powered by battery is the simplest one among all vehicles. It must include an Onboard Battery Charger (OBC) and there are several solutions for its configuration.





Full electric vehicles FCEV

- **Fuel Cell Electric Vehicle (FCEV):** The electrical powertrain is the same as for BEV, the difference comes from the energy source (fuel cell). In the figure there is the system used by Toyota Mirai.





Pros and Cons Hybrid Configurations



Pros and Cons

P0 configuration

Main advantage, this is the most cost to effectiveness ratio configuration:

- Limited impact of the 48V system on the existing vehicle architecture;
- Vehicle and transmission architecture can be kept the same as for a conventional vehicle;
- The 48V electric machine can be easily integrated into the already existing ICE accessories belt drive replacing the 12V alternator (generator);
- In one word: **Cheap**

Disadvantages:

- Traditional engine and electric machine mechanically linked through the accessory belt, so engine friction will be a parasitic loss for the electric machine when it gives boost torque and when it's recuperating electrical energy;
- Belt durability needs to be increased to sustain higher torque and more engine off/on cycles;
- Belt slip can limit the amount of transmitted torque, so current BSG applications still use the 12V starter for cold engine start that demands high electric machine torque due to high engine friction.



Pros and Cons

P1 configuration

Main advantage, the removal of the belt drive:

- Slight increase in efficiency;
- Higher electric machine torque in terms of amplitude and response time (compared to P0);
- Electric creeping possible;

Disadvantages:

- Higher cost;
- Higher impact on existing vehicle architecture (crankshaft has to be changed, integration require re-design of the engine allocation);
- Electric machine is still on ICE side (no mechanical disconnection possible).



Pros and Cons

P2 to P4 configuration

Main advantage, electric machine mounted on driveline side (EM can be detached from ICE):

- Increased energy recuperation potential;
- Very high increase in torque boosting efficiency (moving from P2 to P4, less part to move);
- Electric creep or even electric drive;
- Higher range of coasting strategies;
- Electric 4WD (P4 only)

Disadvantages:

- High impact on existing vehicle architecture;
- Higher overall cost of the electrical components;
- Air cooling not possible for the electric machine;
- S&S inhibited.



EM

Pros and Cons



Pros and Cons

Induction Motor

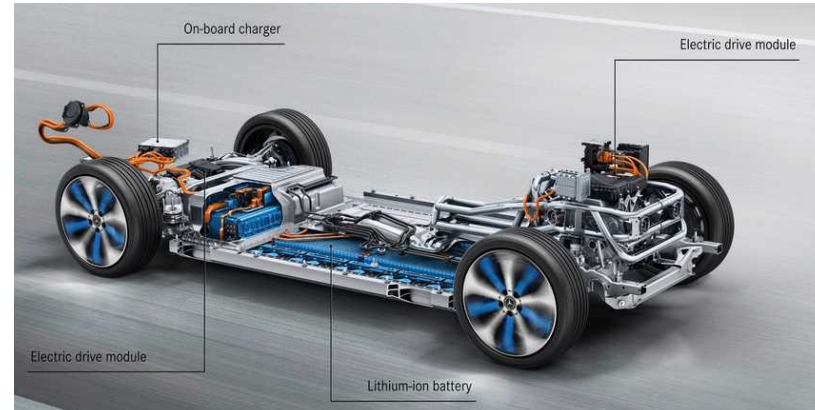
PROS:

- Simple and robust design
- Low cost
- High efficiency at low to medium speeds
- Low maintenance requirements

CONS:

- Poor power density
- Poor performance at high speeds

2020 Mercedes-Benz EQC: 2 induction motors





Pros and Cons

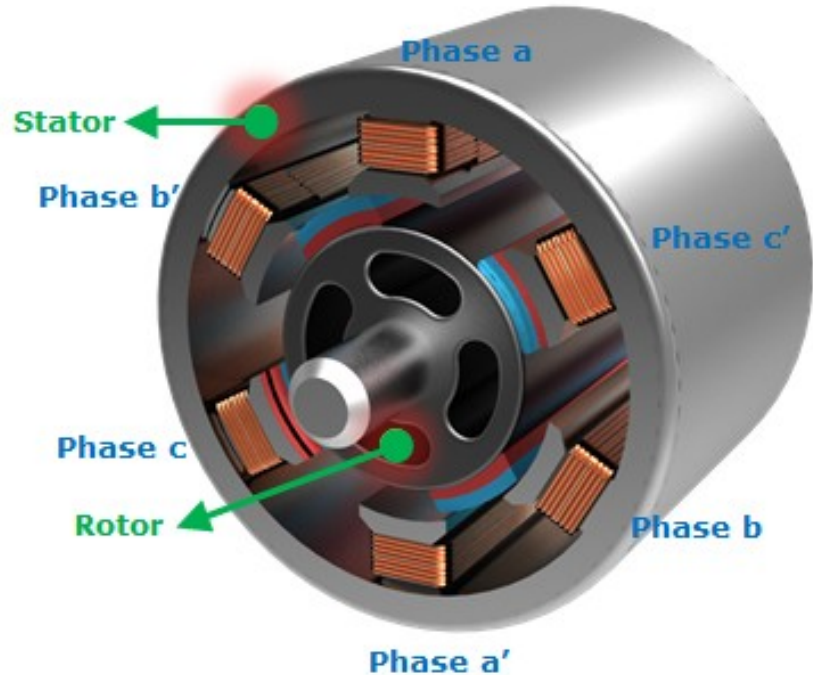
PMSM and BLDC

PROS:

- High power density
- High efficiency
- High torque density
- Responsiveness
- Maintenance-free

CONS:

- High costs
- More complex control
- Sensitive to temperature





Pros and Cons

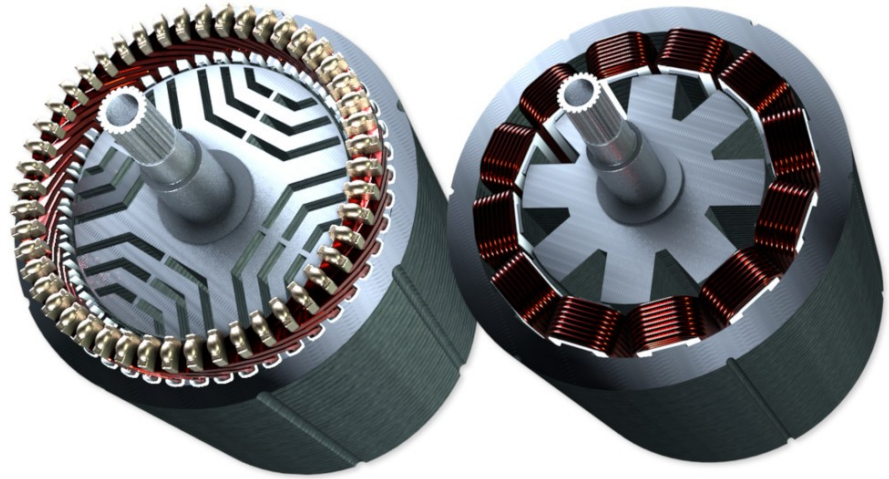
SRM

PROS:

- Simple design
- High reliability
- Low cost
- Good efficiency
- High power density

CONS:

- Noisier than other motor types
- Poor speed control
- Lower power factor
- Higher torque ripple





HEV

Advantages and Disadvantages

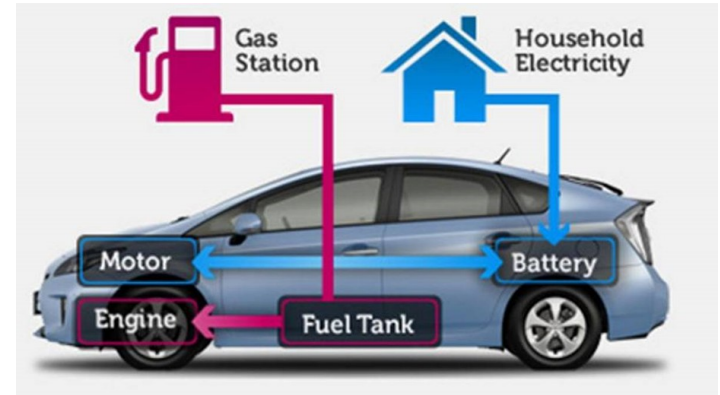


Advantages and disadvantages Hybrid

How convenient are hybrid cars?

Hybrid cars ensure various benefits, specifically for those who do not want or cannot opt for pure electric.

- The main advantages of hybrid cars are:
- Wide range of models to choose from;
- Different hybrid technologies for every need;
- Reduced emissions and consumption compared to only ICE ones;
- No limitation of range compared to electric;
- Various benefits available.





Advantages and disadvantages

Hybrid vs Electric

The real question that most people are asking now is: Is electric or hybrid more convenient?

Electric cars are made up of an electric powertrain that ensures zero emissions, therefore they allow you to move without polluting. Despite this, some negative aspects must be considered: the costs of a completely electric reality are still very high, the limitation linked to autonomy and the need to plan each journey based on the battery recharge needs.

The hybrid car is more suitable for those who have to travel many kilometers a year, in this way it is possible to benefit from greater autonomy, while still reducing carbon dioxide emissions and average consumption.

The hybrid is also the best solution for those who do not have re-charging infrastructures, in fact in the absence of electric stations it is still possible to circulate in petrol mode.

Hybrid vehicles can also be advantageous for those who need more flexibility, for example for driving both on the motorway and in the city. In this case it is possible to take advantage of the performance of the internal combustion engine in the extra-urban context, while in the urban one with the plug-in hybrid or full hybrid it is possible to travel with zero emissions in the city up to 40-60 km.



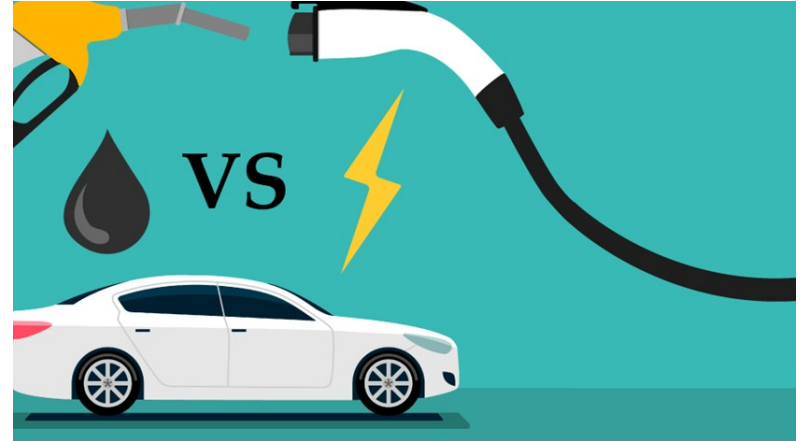
Advantages and disadvantages

Hybrid vs LPG

Is it better hybrid car or LPG?

Quite a doubt, however, concerns the convenience of a hybrid car compared to a LPG car. In fact, gas is a fuel with an advantageous price, as it costs about half compared to petrol. However, it is also more polluting, in addition to the costs for the system, including the mandatory review of the cylinder and replacement after 10 years.

Naturally there is no single answer, in any case a LPG vehicle is convenient for those who travel many kilometers in the city, or as an alternative to diesel cars. Hybrid cars are suitable for those who want a more versatile car that generally lasts longer, moreover it is a more ecological mobility option than LPG and versatile for those who travel often even in the extra-urban context.





Advantages and disadvantages

Hybrid on highway

Is a hybrid car worth it on the highway?

The efficiency of hybrid cars is reduced on the highway, in fact in these situations the electric motor is not very effective, as the petrol or diesel one is mainly used. The absence of electric powertrain support does not allow to maintain low consumption and emissions, with worse performance than driving in the city and on extra-urban roads.

For those who drive mainly on the motorway, the most suitable option is not the hybrid, but it is better to opt for pure electric or highly efficient diesel cars. The former also allow you to travel on the motorway network without polluting, despite the fact that the infrastructure is not yet widespread, the latter ensure good consumption and low emissions thanks to the latest generation diesel engines which allow you to save fuel on the motorway



Automotive examples



Example Audi e-tron

LAUNCH:

- **Announced** 2020
- **Status** Available. Released 2020
- **Base price** € 94,800 \$ 86,440 £ 87,710

BATTERY:

- **Capacity** 86.5 kWh usable, 95 kWh total
- **Tech** Liquid-cooled Li-ion 400V
- **Range** 374 km WLTP (335 km EPA)
- **Consumption** 26.1 kWh/100 km WLTP (28.6 kWh/100 km EPA, 27.1 kWh/100 km NEDC)
- **Recuperation** 3 levels, via driving modes, optional steering wheel paddles
- **Heat pump** Yes
- **AC Charging** Type 2 11 kW, 0-100% in 9 hours 15 min
- **DC Charging** CCS 150 kW max, 0-100% in 50 min



PERFORMANCE:

- **Power** AWD 370 kW (503 hp)
- **Torque** 973 Nm (718 lb-ft)
- **Acceleration** 4.5 sec 0-100 km/h
- **Max speed** 210 km/h (130 mph)



Example

Audi e-tron



370kW

All Wheel Drive



4.5sec

0-100 km/h



95kWh

Nominal capacity

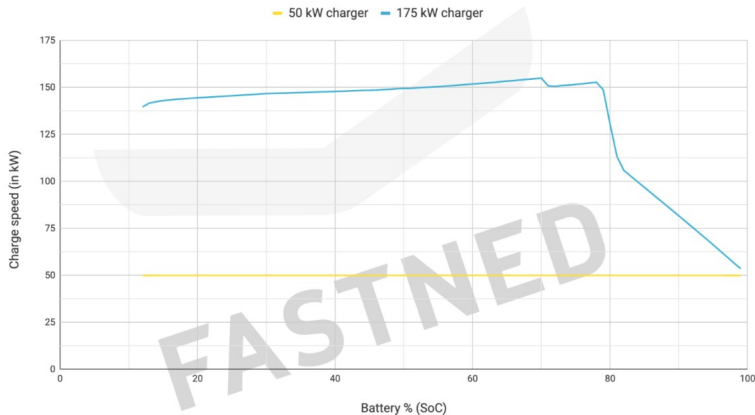


374km

WLTP range

CHARGING CURVE

Audi e-tron Quattro 55



CHARGING TIME

Target charging range:

10% 80%

Charger type:

150 kW DC

0:27h



Audi e-tron S 2019 -



Example Mercedes EQC 400

LAUNCH:

- **Announced** 2021, February
- **Status** Available. Released 2021, Q2
- **Base price** € 66,070 £ 66,405

BATTERY:

- **Capacity** 80 kWh usable, 84.5 kWh total
- **Tech** Air-cooled Li-ion 400V
- **Range** 462 km WLTP (471 km NEDC)
- **Consumption** 21.5 kWh/100 km WLTP
- **Recuperation** 5 levels, via shift paddles
- **Heat pump** Yes
- **AC Charging** Type 2 11 kW, 10-100% in 7 hours 30 min
- **DC Charging** CCS 110 kW max, 10-80% in 40 min



PERFORMANCE:

- **Power** AWD 300 kW (408 hp)
- **Torque** 760 Nm (561 lb-ft)
- **Acceleration** 5.1 sec 0-100 km/h
- **Max speed** 180 km/h (112 mph)



Example Mercedes EQC 400



300kW
All Wheel Drive



5.1sec
0-100 km/h



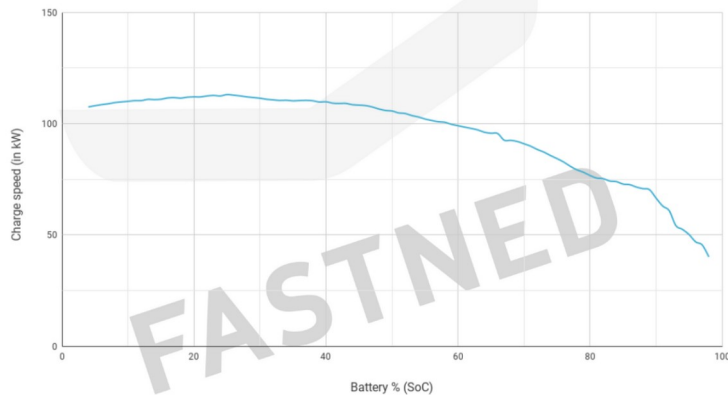
80kWh
Usable capacity



462km
WLTP range

CHARGING CURVE

Mercedes-Benz EQB



CHARGING TIME

Target charging range:

10%

80%

Charger type:

150 kW DC

0:36h



Mercedes EQC 400 4MATIC 2021 -



Example Tesla Model S LR



415kW

All Wheel Drive



3.2sec

0-100 km/h



100kWh

Nominal capacity



634km

WLTP range



LAUNCH:

- **Announced** 2021
- **Status** Available. Released 2021
- **Base price** \$ 95,000

PERFORMANCE:

- **Power** AWD 415 kW (670 hp)
- **Torque** 755 Nm (557 lb-ft)
- **Acceleration** 3.2 sec 0-100 km/h (3.1 sec 0-60 mph)
- **Max speed** 250 km/h (155 mph)

BATTERY:

- **Capacity** 100 kWh total
- **Tech** Liquid-cooled Li-ion 350V
- **Range** 634 km WLTP (652 km EPA)
- **Consumption** 17.4 kWh/100 km EPA
- **Recuperation** 1 level, no coasting, no regen when pedal braking
- **Heat pump** Yes
- **AC Charging** Type 2 11 kW, 0-100% in 10 hours
- **DC Charging** Supercharger V3 250 kW max, 10-80% in 30 min CCS



Example Porsche Taycan Plus Sport Turismo

LAUNCH:

- **Announced** 2021
- **Status** Available. Released 2022
- **Base price** € 86,500

BATTERY:

- **Capacity** 83.7 kWh usable, 93.4 kWh total
- **Tech** Liquid-cooled Li-ion 800V
- **Range** 492 km WLTP
- **Consumption** 21 kWh/100 km WLTP
- **Recuperation** Level 0 to 3 on steering wheel
- **Heat pump** Yes
- **AC Charging** Type 2 11 kW, 0-100% in 9 hours, Type 2 22 kW, 0-100% in 5 hours
- **DC Charging** CCS 270 kW max, 5-80% in 22 min



PERFORMANCE:

- **Power** RWD 300 kW (408 hp)
- **Torque** 345 Nm (254 lb-ft)
- **Acceleration** 5.4 sec 0-100 km/h
- **Max speed** 230 km/h (130 mph)



Example Audi e-tron GT quattro

LAUNCH:

- **Announced** 2021
- **Status** Available. Released 2021
- **Base price** € 103,650 £ 82,150

BATTERY:

- **Capacity** 3.7 kWh usable, 93.4 kWh total
- **Tech** Liquid-cooled Li-Ion 800V
- **Range** 488 km WLTP
- **Consumption** 19.9 kWh/100 km WLTP (18.8 kWh/100 km NEDC)
- **Recuperation** 3 levels, via driving modes, optional steering wheel paddles
- **Heat pump** Yes
- **AC Charging** Type 2 11 kW, 0-100% in 9 hours 15 min, Type 2 22 kW, 0-100% in 4 hours 45 min
- **DC Charging** CCS 800V 270 kW max, 10-80% in 17 min



PERFORMANCE:

- **Power** AWD 390 kW (530 hp)
- **Torque** 640 Nm (472 lb-ft)
- **Acceleration** 4.1 sec 0-100 km/h
- **Max speed** 245 km/h (152 mph)



Example

Audi e-tron GT quattro



390kW

All Wheel Drive



4.1sec

0-100 km/h



93.4kWh

Nominal capacity

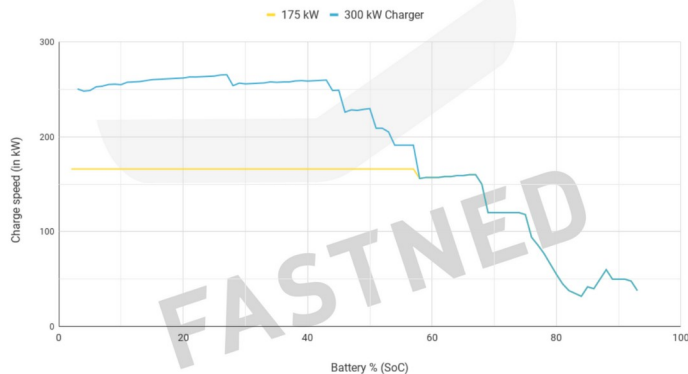


488km

WLTP range

CHARGING CURVE

Audi e-tron GT



CHARGING TIME

Target charging range:

10% 80%

Charger type:

150 kW DC

0:27h



Audi e-tron GT quattro 2021 -



Example Renault Megan E-Tech



96kW

Front Wheel Drive



10sec

0-100 km/h



40kWh

Usable capacity



298km

WLTP range



LAUNCH:

- **Announced** 2021
- **Status** Available. Released 2022, February
- **Base price** € 42,000

PERFORMANCE:

- **Power** FWD 96 kW (130 hp)
- **Torque** 250 Nm (184 lb-ft)
- **Acceleration** 10 sec 0-100 km/h
- **Max speed** 150 km/h (93 mph)

BATTERY:

- **Capacity** 40 kWh usable
- **Tech** Liquid-cooled Li-ion 352V
- **Range** 298 km WLTP
- **Consumption** 15.7 kWh/100 km WLTP
- **Recuperation** Level 0 to 3 on shift paddles
- **Heat pump** Yes
- **AC Charging** Type 2 7 kW, Type 2 22 kW
- **DC Charging** CCS 85 kW max, 200 km in 30 min (the version initially was released without fast charging)



Example

BMW iX3

LAUNCH:

- **Announced** 2021, September
- **Status** Available. Released 2021, September
- **Base price** € 67,300 £ 60,970

BATTERY:

- **Capacity** 74 kWh usable, 80 kWh total
- **Tech** Air-cooled Li-ion 400V
- **Range** 453 km WLTP (520 km NEDC)
- **Consumption** 18.9 kWh/100 km WLTP
- **Recuperation** 4 modes, via gear lever
- **Heat pump** Yes
- **AC Charging** Type 2 11 kW, 0-100% in 7 hours 30 min
- **DC Charging** CCS 150 kW max, 10-80% in 34 min



PERFORMANCE:

- **Power** RWD 210 kW (286 hp)
- **Torque** 400 Nm (295 lb-ft)
- **Acceleration** 6.8 sec 0-100 km/h
- **Max speed** 180 km/h (112 mph)



Example

BMW iX3



210kW

Rear Wheel Drive



6.8sec

0-100 km/h



74kWh

Usable capacity

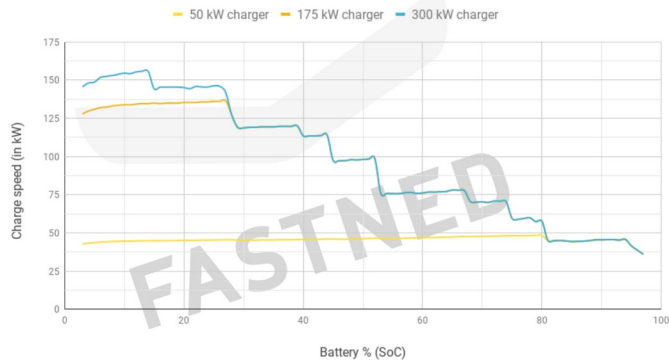


453km

WLTP range

CHARGING CURVE

BMW iX3



CHARGING TIME

Target charging range:

10% 80%

Charger type:

150 kW DC

0:34h



BMW iX3 2021 -



Final Comparison

E-Motors

TYPE	INDUCTION MOTOR	PMSM/IPM ⁽²⁾	SM ⁽³⁾	SRM ⁽⁴⁾
Peak efficiency	>90%	>94%	>90%	>92%
Max. speed	+	O	+	++
Spec. Power ⁽¹⁾ [kW/kg]	~ 1,0	~ 1,6	< 1,0	~ 1,0
Manufacturing costs	-	+	o	--
Technoogy maturity	++	+	+	o
Failure torque	None	++	-	None
Noise/vibration	-	-	-	+
Automotive application	✓	✓	✓	?
Key reasons	Low costs	Low weight, Compact design	No permanent magnets	Low costs

– Very low - Low o Medium + High ++ Very high

(1) Permanent power (2) Permanent magnet synchronous motor/Internal permanent magnet motor

(3) Synchronous motor (4) Synchronous reluctance motor

Global Market Developments in Electric Traction Motors, SAE 2010 Powertrain Electric Motors Symposium, USA



Final Comparison

Max Speed

Model	Max speed [km/h]	0-100 [s]	Range [km]	Capacity [kWh]
Tesla ModelS LR	250	3,2	634	100
Audi e-tron GT quattro	245	4,1	488	93,4
Porsche Tycan Plus Sport Turismo	230	5,4	433	79
Tesla ModelY LR	217	6,9	370	50
Audi e-tron	210	4,5	375	95
Jaguar I Pace EV400	200	4,8	470	90
BMW i4	190	5,7	190	84
Hyundai Ioniq 5	185	8,5	400	62
Kia EV6 SR RWD	185	8,5	394	62
Mercedes EQC 400 4MATIC	180	5,1	462	85
BMW iX3	180	6,8	453	80
Fiat 500e	150	9	312	42
Renault Megane E-Tech	150	10	298	40
Smart EQ ForFour	130	12,7	130	17,6



Final Comparison

Fastest 0-100

Model	Max speed [km/h]	0-100 [s]	Range [km]	Capacity [kWh]
Tesla ModelS LR	250	3,2	634	100
Audi e-tron GT quattro	245	4,1	488	93,4
Audi e-tron	210	4,5	375	95
Jaguar I Pace EV400	200	4,8	470	90
Mercedes EQC 400 4MATIC	180	5,1	462	85
Porsche Tycan Plus Sport Turismo	230	5,4	433	79
BMW i4	190	5,7	190	84
BMW iX3	180	6,8	453	80
Tesla ModelY LR	217	6,9	370	50
Hyundai Ioniq 5	185	8,5	400	62
Kia EV6 SR RWD	185	8,5	394	62
Fiat 500e	150	9	312	42
Renault Megane E-Tech	150	10	298	40
Smart EQ ForFour	130	12,7	130	17,6



Final Comparison

Longest Range

Model	Max speed [km/h]	0-100 [s]	Range [km]	Capacity [kWh]
Tesla ModelS LR	250	3,2	634	100
Audi e-tron GT quattro	245	4,1	488	93,4
Jaguar I Pace EV400	200	4,8	470	90
Mercedes EQC 400 4MATIC	180	5,1	462	85
BMW iX3	180	6,8	453	80
Porsche Tycan Plus Sport Turismo	230	5,4	433	79
Hyundai Ioniq 5	185	8,5	400	62
Kia EV6 SR RWD	185	8,5	394	62
Audi e-tron	210	4,5	375	95
Tesla ModelY LR	217	6,9	370	50
Fiat 500e	150	9	312	42
Renault Megane E-Tech	150	10	298	40
BMW i4	190	5,7	190	84
Smart EQ ForFour	130	12,7	130	17,6



Final Comparison

Higher Battery Capacity

Model	Max speed [km/h]	0-100 [s]	Range [km]	Capacity [kWh]
Tesla ModelS LR	250	3,2	634	100
Audi e-tron	210	4,5	375	95
Audi e-tron GT quattro	245	4,1	488	93,4
Jaguar I Pace EV400	200	4,8	470	90
Mercedes EQC 400 4MATIC	180	5,1	462	85
BMW i4	190	5,7	190	84
BMW iX3	180	6,8	453	80
Porsche Tycan Plus Sport Turismo	230	5,4	433	79
Hyundai Ioniq 5	185	8,5	400	62
Kia EV6 SR RWD	185	8,5	394	62
Tesla ModelY LR	217	6,9	370	50
Fiat 500e	150	9	312	42
Renault Megane E-Tech	150	10	298	40
Smart EQ ForFour	130	12,7	130	17,6

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